

PROJECT : 3

SECURE MEDIA HOSTING PLATFORM

Launch EC2 instance with restricted ssh access

Create private S3 bucket

Upload media files to S3

Server media via EC2 web server

➤ Launch EC2 instance

The screenshot displays the AWS Management Console interface for an EC2 instance. The browser address bar shows the URL: `eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#InstanceDetails:instanceId=i-0221acc050f516380`. The console header includes the AWS logo, a search bar, and navigation links for EC2, Instances, and the specific instance ID. The left sidebar contains a navigation menu with categories like EC2, Images, and Elastic Block Store. The main content area is titled "Instance summary for i-0221acc050f516380 (webserver)" and shows the instance is in a "Running" state. Key details include the Public IPv4 address (51.20.118.87), Private IPv4 address (172.31.27.23), and Public DNS (ec2-51-20-118-87.eu-north-1.compute.amazonaws.com). The instance type is t3.micro, and it is associated with VPC ID vpc-0b70c516bf88cd647 and Subnet ID subnet-0f8b0a5b800183eb8. The instance ARN is `arn:aws:ec2:eu-north-1:014576993102:instance/i-0221acc050f516380`. The console also shows an AWS Compute Optimizer finding recommending to opt-in for recommendations.

Instance summary for i-0221acc050f516380 (webserver) Info

Updated less than a minute ago

Instance ID i-0221acc050f516380	Public IPv4 address 51.20.118.87 open address	Private IPv4 addresses 172.31.27.23
IPv6 address -	Instance state Running	Public DNS ec2-51-20-118-87.eu-north-1.compute.amazonaws.com open address
Hostname type IP name: ip-172-31-27-23.eu-north-1.compute.internal	Private IP DNS name (IPv4 only) ip-172-31-27-23.eu-north-1.compute.internal	Elastic IP addresses -
Answer private resource DNS name IPv4 (A)	Instance type t3.micro	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn more
Auto-assigned IP address 51.20.118.87 [Public IP]	VPC ID vpc-0b70c516bf88cd647	Auto Scaling Group name -
IAM role -	Subnet ID subnet-0f8b0a5b800183eb8	Managed false
IMDSv2 Required	Instance ARN arn:aws:ec2:eu-north-1:014576993102:instance/i-0221acc050f516380	
Operator		



Connect EC2 instance and install httpd

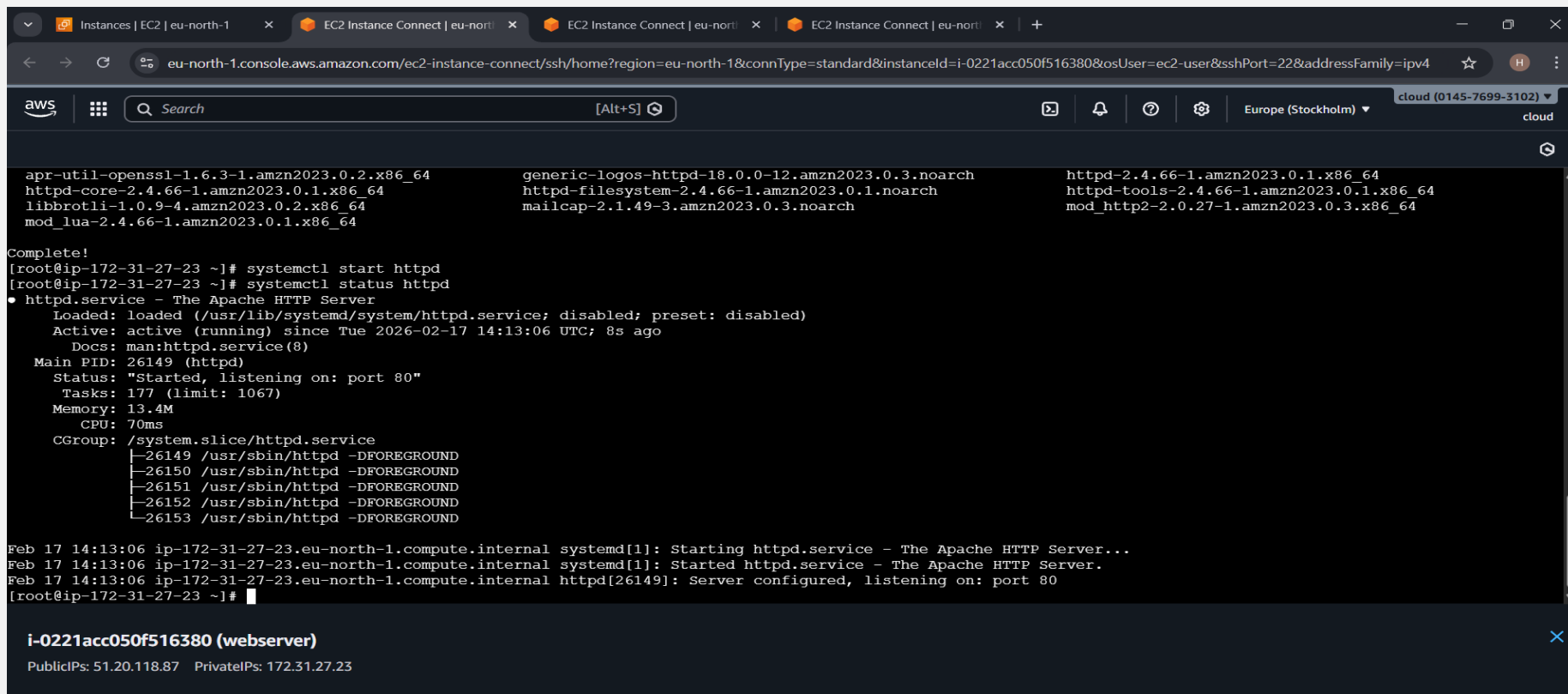
The screenshot shows the AWS Management Console interface with multiple tabs for EC2 Instance Connect sessions. The active session shows a terminal window for an Amazon Linux 2023 instance. The terminal output shows the user running `sudo -i` to become root, then `yum install httpd -y` to install the httpd package. The output shows the package list and the installation progress.

```
[ec2-user@ip-172-31-27-23 ~]$ sudo -i
[root@ip-172-31-27-23 ~]# yum install httpd -y
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
245 kB/s | 31 kB    00:00
```

Package	Architecture	Version	Repository	Size
Installing:				
httpd	x86_64	2.4.66-1.amzn2023.0.1	amazonlinux	47 k
Installing dependencies:				
apr	x86_64	1.7.5-1.amzn2023.0.4	amazonlinux	129 k
apr-util	x86_64	1.6.3-1.amzn2023.0.2	amazonlinux	97 k
apr-util-ldap	x86_64	1.6.3-1.amzn2023.0.2	amazonlinux	13 k
generic-logos-httpd	noarch	18.0.0-12.amzn2023.0.3	amazonlinux	19 k
httpd-core	x86_64	2.4.66-1.amzn2023.0.1	amazonlinux	1.4 M
httpd-filesystem	noarch	2.4.66-1.amzn2023.0.1	amazonlinux	13 k
httpd-tools	x86_64	2.4.66-1.amzn2023.0.1	amazonlinux	81 k
libbrotli	x86_64	1.0.9-4.amzn2023.0.2	amazonlinux	315 k

i-0221acc050f516380 (webserver)
PublicIPs: 51.20.118.87 PrivateIPs: 172.31.27.23

➤ HERE WE USE COMMAND SYSTEMCTL TO START AND CHECK STATUS OF HTTPD



```
apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64      generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch  httpd-2.4.66-1.amzn2023.0.1.x86_64
httpd-core-2.4.66-1.amzn2023.0.1.x86_64          httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch      httpd-tools-2.4.66-1.amzn2023.0.1.x86_64
libbrotli-1.0.9-4.amzn2023.0.2.x86_64            mailcap-2.1.49-3.amzn2023.0.3.noarch              mod_http2-2.0.27-1.amzn2023.0.3.x86_64
mod_lua-2.4.66-1.amzn2023.0.1.x86_64

Complete!
[root@ip-172-31-27-23 ~]# systemctl start httpd
[root@ip-172-31-27-23 ~]# systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
   Active: active (running) since Tue 2026-02-17 14:13:06 UTC; 8s ago
     Docs: man:httpd.service(8)
   Main PID: 26149 (httpd)
   Status: "Started, listening on: port 80"
    Tasks: 177 (limit: 1067)
   Memory: 13.4M
      CPU: 70ms
   CGroup: /system.slice/httpd.service
           └─26149 /usr/sbin/httpd -DFOREGROUND
             └─26150 /usr/sbin/httpd -DFOREGROUND
               └─26151 /usr/sbin/httpd -DFOREGROUND
                 └─26152 /usr/sbin/httpd -DFOREGROUND
                   └─26153 /usr/sbin/httpd -DFOREGROUND

Feb 17 14:13:06 ip-172-31-27-23.eu-north-1.compute.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Feb 17 14:13:06 ip-172-31-27-23.eu-north-1.compute.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Feb 17 14:13:06 ip-172-31-27-23.eu-north-1.compute.internal httpd[26149]: Server configured, listening on: port 80
[root@ip-172-31-27-23 ~]#
```

i-0221acc050f516380 (webserver)
PublicIPs: 51.20.118.87 PrivateIPs: 172.31.27.23

➤ We create s3 bucket as private

The screenshot displays the AWS Management Console interface for S3 buckets in the eu-north-1 region. The top navigation bar includes the AWS logo, a search bar, and the current region (Europe (Stockholm)). The breadcrumb trail shows 'Amazon S3 > Buckets'. The main content area is divided into two tabs: 'General purpose buckets' (selected) and 'Directory buckets'. Under 'General purpose buckets', there is a sub-header 'General purpose buckets (1) Info' and a description 'Buckets are containers for data stored in S3.'. Below this is a search bar 'Find buckets by name' and a table with one bucket. The table has columns for 'Name', 'AWS Region', and 'Creation date'. The bucket listed is 'webservice-bucket20' in the 'Europe (Stockholm) eu-north-1' region, created on 'February 17, 2026, 19:45:43 (UTC+05:30)'. To the right of the table are buttons for 'Copy ARN', 'Empty', 'Delete', and 'Create bucket'. On the right side of the console, there are two side panels: 'Account snapshot' (updated daily) with a 'View dashboard' button, and 'External access summary' (updated daily) with a description of external access findings.

Instances | EC2 | eu-north-1 | EC2 Instance Connect | eu-north-1 | S3 buckets | S3 | eu-north-1

eu-north-1.console.aws.amazon.com/s3/buckets?region=eu-north-1

aws Search [Alt+S]

Europe (Stockholm) cloud (0145-7699-3102) cloud

Amazon S3 > Buckets

General purpose buckets All AWS Regions Directory buckets

General purpose buckets (1) Info

Buckets are containers for data stored in S3.

Find buckets by name

Name	AWS Region	Creation date
webservice-bucket20	Europe (Stockholm) eu-north-1	February 17, 2026, 19:45:43 (UTC+05:30)

Copy ARN Empty Delete Create bucket

Account snapshot Info View dashboard

Updated daily

Storage Lens provides visibility into storage usage and activity trends.

External access summary Info

Updated daily

External access findings help you identify bucket permissions that allow public access or access from other AWS accounts.

➤ Block public access and this bucket is fully private

Instances | EC2 | eu-north-1 | EC2 Instance Connect | eu-north-1 | Create S3 bucket | S3 | eu-north-1

eu-north-1.console.aws.amazon.com/s3/bucket/create?region=eu-north-1

Search [Alt+S]

Europe (Stockholm) cloud (0145-7699-3102)

Amazon Services > buckets > Create bucket

Object Ownership [Info](#)

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

Object Ownership

☒ **ACLs disabled (recommended)**
All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.

☐ **ACLs enabled**
Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

Object Ownership

Bucket owner enforced

Block Public Access settings for this bucket

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

☒ **Block all public access**
Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

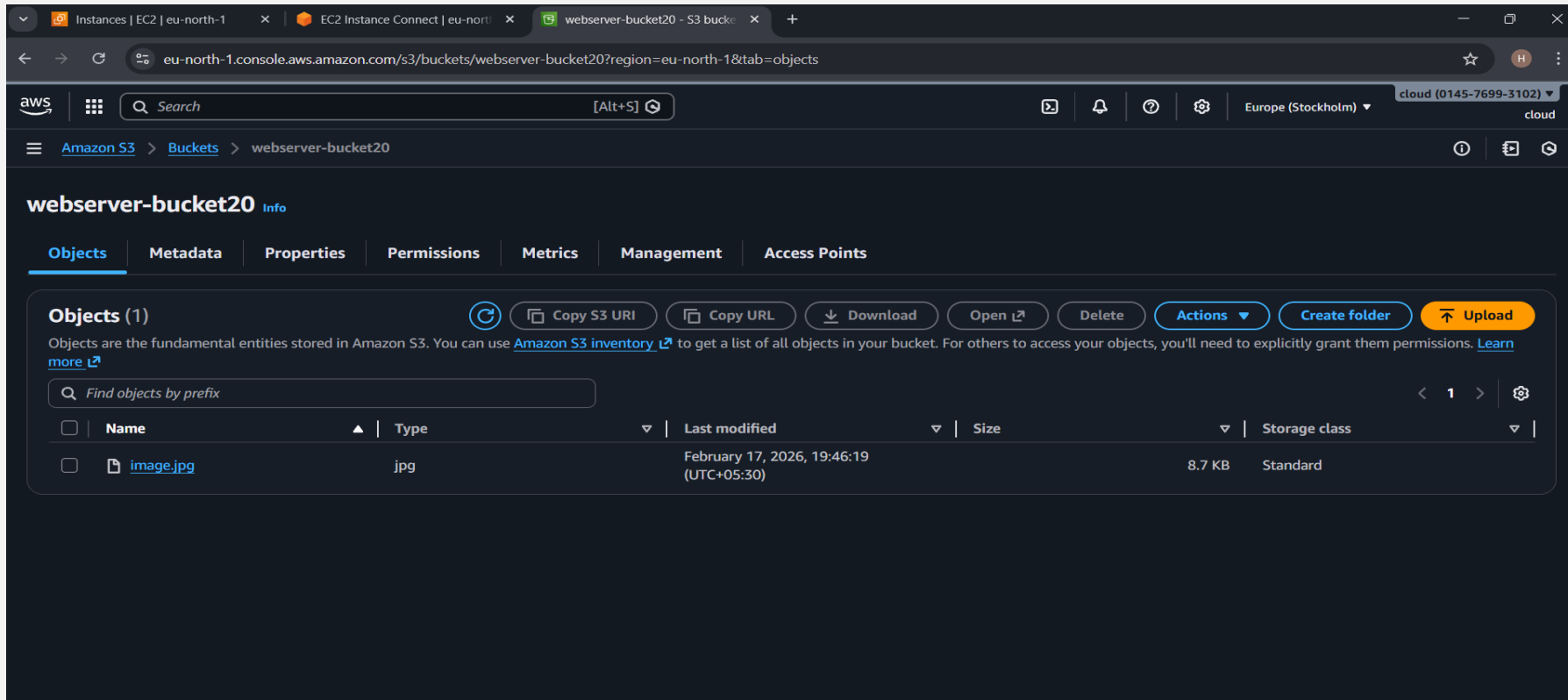
☒ **Block public access to buckets and objects granted through *new* access control lists (ACLs)**
S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.

☒ **Block public access to buckets and objects granted through *any* access control lists (ACLs)**
S3 will ignore all ACLs that grant public access to buckets and objects.

☒ **Block public access to buckets and objects granted through *new* public bucket or access point policies**
S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.

☒ **Block public and cross-account access to buckets and objects through *any* public bucket or access point policies**
S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

➤ Upload media files to S3



The screenshot displays the AWS Management Console interface for the 'webserver-bucket20' S3 bucket. The breadcrumb navigation shows 'Amazon S3 > Buckets > webserver-bucket20'. The bucket's name 'webserver-bucket20' is prominently displayed at the top, with an 'Info' link. Below this, a series of tabs allows navigation between different views: 'Objects' (selected), 'Metadata', 'Properties', 'Permissions', 'Metrics', 'Management', and 'Access Points'. The 'Objects' tab shows a list of objects, currently containing one item: 'image.jpg'. Above the list, there are several action buttons: 'Copy S3 URI', 'Copy URL', 'Download', 'Open', 'Delete', 'Actions' (with a dropdown arrow), 'Create folder', and 'Upload'. A search bar labeled 'Find objects by prefix' is positioned above the table. The table itself has columns for selection, Name, Type, Last modified, Size, and Storage class. The object 'image.jpg' is listed with a last modified date of February 17, 2026, at 19:46:19 (UTC+05:30), a size of 8.7 KB, and a storage class of Standard.

Instances | EC2 | eu-north-1 | EC2 Instance Connect | eu-north-1 | webserver-bucket20 - S3 bucket

eu-north-1.console.aws.amazon.com/s3/buckets/webserver-bucket20?region=eu-north-1&tab=objects

Search [Alt+S]

Europe (Stockholm) cloud (0145-7699-3102) cloud

Amazon S3 > Buckets > webserver-bucket20

webserver-bucket20 Info

Objects Metadata Properties Permissions Metrics Management Access Points

Objects (1)   Copy S3 URI  Copy URL  Download  Open  Delete  Actions  Create folder  Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

☐	Name	Type	Last modified	Size	Storage class
☐	 image.jpg	jpg	February 17, 2026, 19:46:19 (UTC+05:30)	8.7 KB	Standard

- Give EC2 access to s3 (secure way –IAM role)
we create role for s3 full access

The screenshot displays the AWS IAM console interface. The left-hand navigation pane shows the 'IAM' menu expanded, with 'Roles' selected. The main content area is titled 'access-role' and includes a description: 'Allows EC2 instances to call AWS services on your behalf.' Below this, a 'Summary' section provides key details:

- Creation date:** February 17, 2026, 19:48 (UTC+05:30)
- ARN:** `arn:aws:iam::014576993102:role/access-role`
- Instance profile ARN:** `arn:aws:iam::014576993102:instance-profile/access-role`
- Last activity:** -
- Maximum session duration:** 1 hour

Below the summary, there are tabs for 'Permissions', 'Trust relationships', 'Tags', 'Last Accessed', and 'Revoke sessions'. The 'Permissions' tab is active, showing a section for 'Permissions policies' with a search bar and a 'Filter by Type' dropdown set to 'All types'. At the bottom, a table header is visible with columns for 'Policy name', 'Type', and 'Attached entities'. The table is currently empty, with a 'Loading policies' message at the bottom.



We attach IAM role to ec2 instance

Modify IAM role | EC2 | eu-north-1 | EC2 Instance Connect | eu-north-1 | webserver-bucket20 - S3 bucket | access-role | IAM | Global

eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#ModifyIAMRole:instanceId=i-0221acc050f516380

aws Search [Alt+S] Ask Amazon Q Europe (Stockholm) cloud (0145-7699-3102) cloud

EC2 > Instances > i-0221acc050f516380 > Modify IAM role

Modify IAM role [Info](#)

Attach an IAM role to your instance.

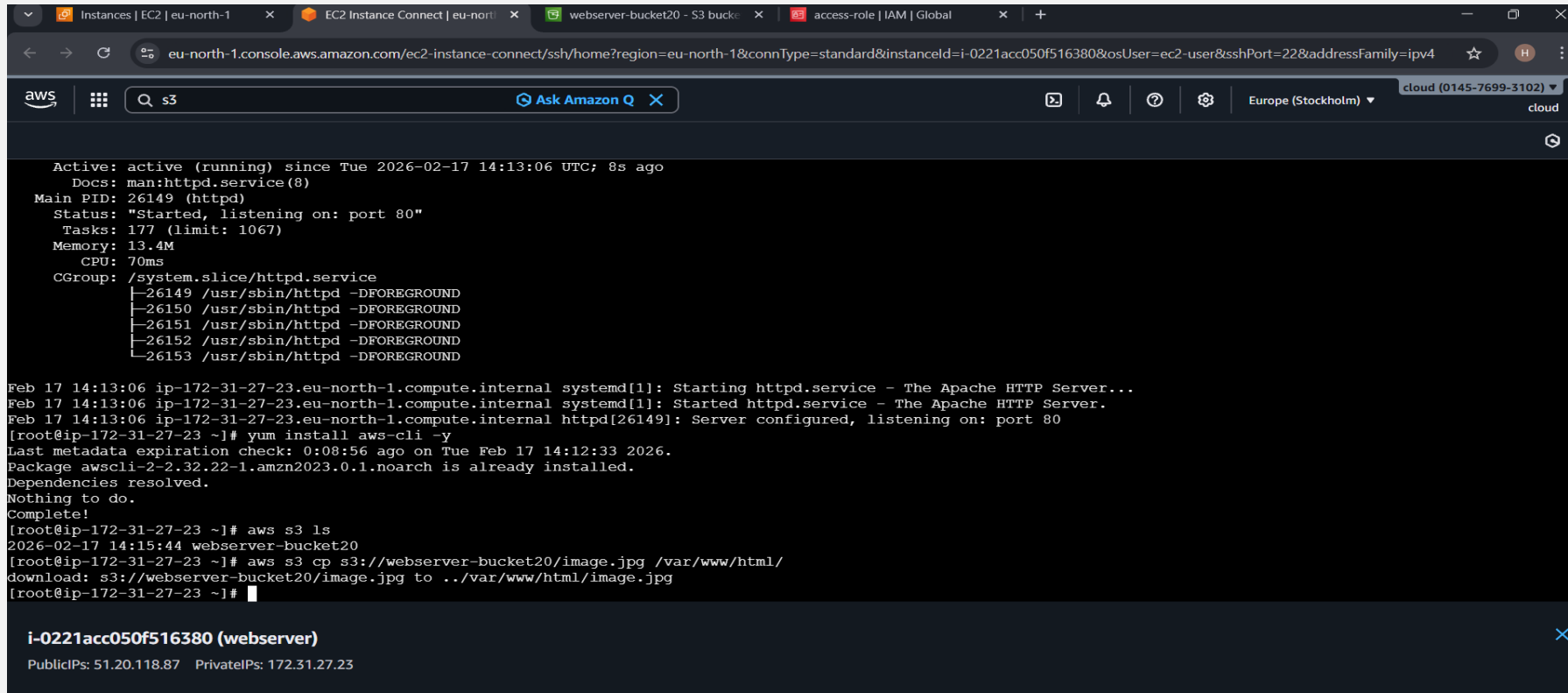
Instance ID
i-0221acc050f516380 (webserver)

IAM role
Select an IAM role to attach to your instance or create a new role if you haven't created any. The role you select replaces any roles that are currently attached to your instance.

access-role ▼ [Create new IAM role](#)

[Cancel](#) [Update IAM role](#)

➤ Install httpd and aws cli on EC2 instance



The screenshot shows a terminal window within the AWS Management Console, connected to an EC2 instance named 'webserver-bucket20'. The terminal output displays the status of the httpd service, which is active and listening on port 80. It also shows the successful installation of the aws-cli package using yum. Finally, it demonstrates the upload of a file from an S3 bucket to the instance's local file system.

```
Active: active (running) since Tue 2026-02-17 14:13:06 UTC; 8s ago
  Docs: man:httpd.service(8)
Main PID: 26149 (httpd)
Status: "Started, listening on: port 80"
Tasks: 177 (limit: 1067)
Memory: 13.4M
CPU: 70ms
CGroup: /system.slice/httpd.service
└─26149 /usr/sbin/httpd -DFOREGROUND
└─26150 /usr/sbin/httpd -DFOREGROUND
└─26151 /usr/sbin/httpd -DFOREGROUND
└─26152 /usr/sbin/httpd -DFOREGROUND
└─26153 /usr/sbin/httpd -DFOREGROUND

Feb 17 14:13:06 ip-172-31-27-23.eu-north-1.compute.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Feb 17 14:13:06 ip-172-31-27-23.eu-north-1.compute.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Feb 17 14:13:06 ip-172-31-27-23.eu-north-1.compute.internal httpd[26149]: Server configured, listening on: port 80
[root@ip-172-31-27-23 ~]# yum install aws-cli -y
Last metadata expiration check: 0:08:56 ago on Tue Feb 17 14:12:33 2026.
Package awscli-2-2.32.22-1.amzn2023.0.1.noarch is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@ip-172-31-27-23 ~]# aws s3 ls
2026-02-17 14:15:44 webserver-bucket20
[root@ip-172-31-27-23 ~]# aws s3 cp s3://webserver-bucket20/image.jpg /var/www/html/
download: s3://webserver-bucket20/image.jpg to ../var/www/html/image.jpg
[root@ip-172-31-27-23 ~]#
```

i-0221acc050f516380 (webserver)
PublicIPs: 51.20.118.87 PrivateIPs: 172.31.27.23

➤ Copy media from s3 to web directory

```
Complete!  
[root@ip-172-31-27-23 ~]# aws s3 ls  
2026-02-17 14:15:44 webserver-bucket20  
[root@ip-172-31-27-23 ~]# aws s3 cp s3://webserver-bucket20/image.jpg /var/www/html/  
download: s3://webserver-bucket20/image.jpg to ../var/www/html/image.jpg  
[root@ip-172-31-27-23 ~]#
```

i-0221acc050f516380 (webserver)

PublicIPs: 51.20.118.87 PrivateIPs: 172.31.27.23

➤ Here we use <http://public-ip/image.Jpg> to display output

